HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Processed raw LiDAR point cloud data, classifying ground, vegetation, and structures using **Trimble Business Center** and generating **DTMs and DSMs** for 5+ client projects.
- Converted processed LiDAR data into client-ready **CAD and GIS formats** including DWG, DXF, and shapefiles, ensuring adherence to project-specific coordinate systems like **ITM (EPSG:2157)**.
- Managed large datasets exceeding 500GB, maintaining organised, version-controlled digital workflows and producing technical reports and processing logs for project archival.
- Carried out rigorous **QA/QC** on all data products to meet project accuracy specifications and client standards, identifying and resolving an average of 15 data anomalies per project.
- Collaborated with drone pilots and survey teams on capture planning and ground control requirements, contributing to the successful execution of **UAV-based LiDAR** surveys for infrastructure inspection projects.
- Developed Python scripts to automate repetitive data processing tasks, reducing manual processing time for point cloud classification by an estimated **30%**.
- Performed production support and monitoring of enterprise applications using **Splunk** and **Dynatrace**, managing SLAs and responding to P1/P2 incidents within a financial-services operations context.

SKILLS

LiDAR Processing – LiDAR data processing, point cloud classification, DTM, DSM, coordinate systems, georeferencing, geoid models, elevation modelling, QA/QC

Geospatial Software – Trimble Business Center, DJI Terra, TerraScan, LP360, Global Mapper, CloudCompare, Autodesk ReCap, Civil 3D, AutoCAD, GIS, CAD

Data Formats & GIS/CAD – DWG, DXF, Civil 3D, LAS/LAZ, GeoTIFF, shapefile, Irish Grid, ITM, OSGB

Drone Technology – UAV-based LiDAR systems, drone operations

Programming & Automation – Python, scripting, automation, data pipelines, ETL

Cloud & Monitoring – AWS, EC2, S3, Lambda, CloudWatch, Splunk, Dynatrace

PROJECTS

Airborne LiDAR Survey Data Processing (HCL Technologies)

- Processed airborne LiDAR datasets for a regional infrastructure delivery programme, generating **digital terrain models (DTMs)** and **digital surface models (DSMs)** for large-scale civil engineering projects.
- Executed data classification tasks within **Trimble Business Center** to identify and segment ground, vegetation, and built structures, enabling detailed site analysis for clients.
- Ensured all deliverables met stringent accuracy specifications through meticulous **QA/QC** procedures and managed large data volumes using organised digital workflows.

Automated Geospatial Data Transformation Pipeline (HCL Technologies)

- Developed a **Python**-based pipeline to automate the conversion of raw LiDAR data (LAS/LAZ) into **GeoTIFF** and **shapefile** formats, reducing manual effort by approximately **40%**.
- Integrated coordinate system transformations to reproject data into **ITM (EPSG:2157)**, ensuring compatibility with client GIS platforms and mapping requirements.
- Implemented error logging and reporting mechanisms, improving the traceability and reliability of the data processing workflow for enterprise customers.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)